GUSTAVO LUJAN CENTENO

gustavo_lujancenteno@student.uml.edu | Shrewsbury, MA | (973) 954-6305 | gustavo160902.github.io

EDUCATION

University of Massachusetts Lowell — Lowell, MA
Bachelor of Science in Computer Engineering — May 2026
GPA: 3.33/4.00

University of Massachusetts Lowell — Lowell, MA
Master of Science in Computer Engineering — In-progress
Graduate GPA: 4.00/4.00

SKILLS

Programming: C++, C#, Python, MATLAB, Go, ARM Assembly, VHDL
Electronics: Circuit Design, Multisim, LTSpice, Oscilloscope, Waveform Generator, Soldering
Tools: Git, Analog Discovery, Arduino IDE, Quartus, SolidWorks, Visual Studio Code
Languages: English, Spanish, Italian

RELEVANT COURSEWORK

Operating Systems · Digital Systems & FPGA Design · Microprocessors & Embedded Systems · Computer Architecture · Network Design · Electronics & Circuit Design · Software Engineering

PROJECTS

Fabrication Machine for Microelectrode Arrays — UML Capstone — Spring 2026

- Contributed to automated MEA fabrication system as part of a multidisciplinary engineering team.
- Worked on hardware-software integration, motion control, and testing of embedded subsystems.
- Performed iterative testing, calibration, and debugging to improve system performance and automation.

LoRa Mesh Network Optimization — UML Research — Spring 2026

- Extended a Python simulation by implementing Grey Wolf Optimizer (GWO) and PSO routing algorithms on LoRa mesh networks.
- Designed test sequences and measured latency, energy consumption, and delivery rate across 5–100 node configurations.
- Validated results on real RAK LoRa hardware running Meshtastic firmware.

FPGA Temperature Sensor — Clock Domain Crossing — UML Project — Fall 2025

- Designed a VHDL system on DE10-Lite reading the on-chip temperature sensor and displaying live readings on seven segment displays.
- Implemented async FIFO with Gray code synchronizers to safely transfer data between 1 MHz ADC and 50 MHz display domains.
- Built a custom VHDL clock divider replacing the deprecated PLL megafunction in Quartus Prime Lite 24.1.

Cooperative Real-Time Scheduler — UML Project — Fall 2025

- Implemented a cooperative real-time task scheduler in C/C++ on an AVR microcontroller using the Arduino Uno board.
- Compared Round-Robin vs Priority scheduling measuring latency, deadline misses, and CPU usage across 3 task types.
- Designed scheduling logic, timing counters, and task execution control for multiple concurrent embedded tasks.

CERTIFICATIONS

IBM – Machine Learning for Data Science Certificate
Methodological Principles for TEFL to Children & Teenagers — Pontificia Universidad Católica del Perú

EXPERIENCE

COVID-19 Volunteer | Framingham Town, Framingham, MA — June 2021 – Nov 2021

- Worked with a volunteer team to set up distribution stations for masks and sanitizers.
- Rotated responsibilities including crowd guidance, supply distribution, and keeping events organized.
- Partnered with local health staff and volunteers to reach vulnerable community groups.